DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Lesson strips

DNA, genes and chromosomes

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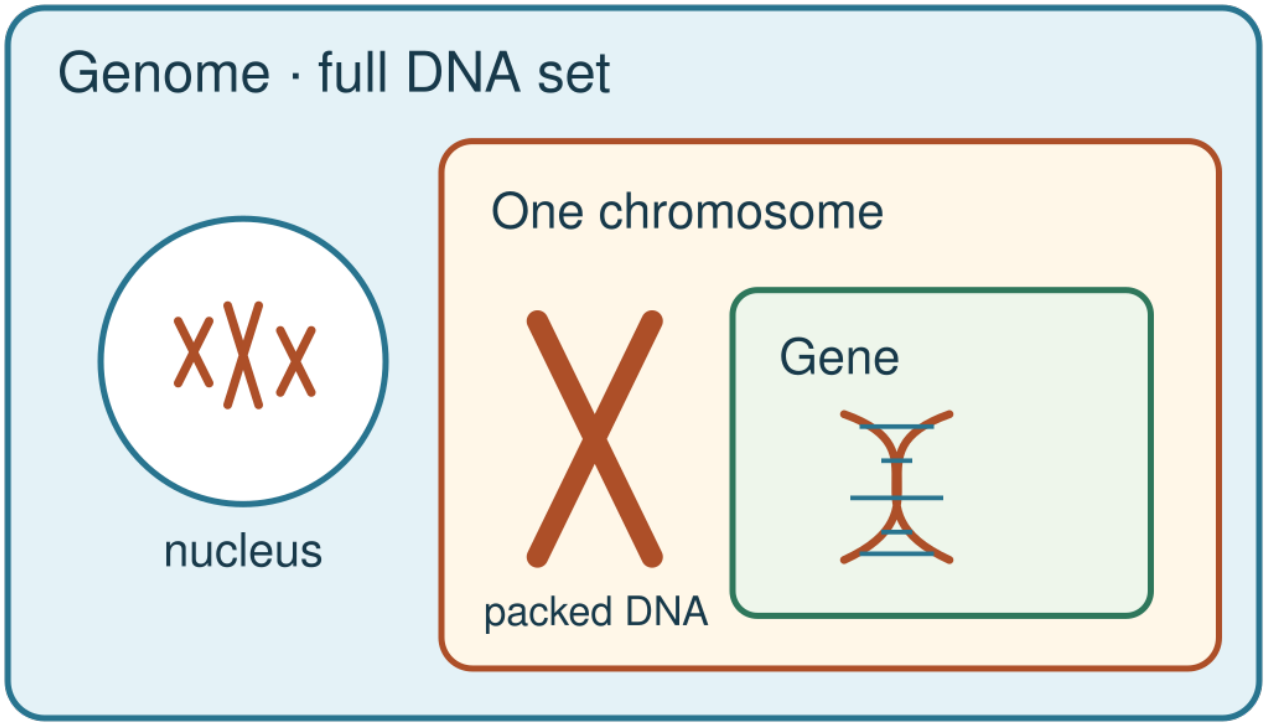
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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Arrival · Supported

Look at this week's string-and-clip DNA model. Point to one detail you notice.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Exit ticket · Supported

1. What does your model show, and why?

Show what you know with a model and explain one reason.

My learning: I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.

How I know: I can show a gene within DNA, DNA packaged in a chromosome, and the genome as all DNA.

2. What feedback did you hear, and what is your next step?

My feedback was … Name what you heard at Review.

Next I need to … Use “section of DNA” and “codes for a protein” precisely in one sentence.

Check today’s success:

I can label DNA, a gene and a chromosome in a simple model.

I can explain that a gene is a section of DNA with a protein instruction.

I can distinguish one chromosome from the genome.

Your teacher hears your answer and uses it to choose our next model, prompt or example.

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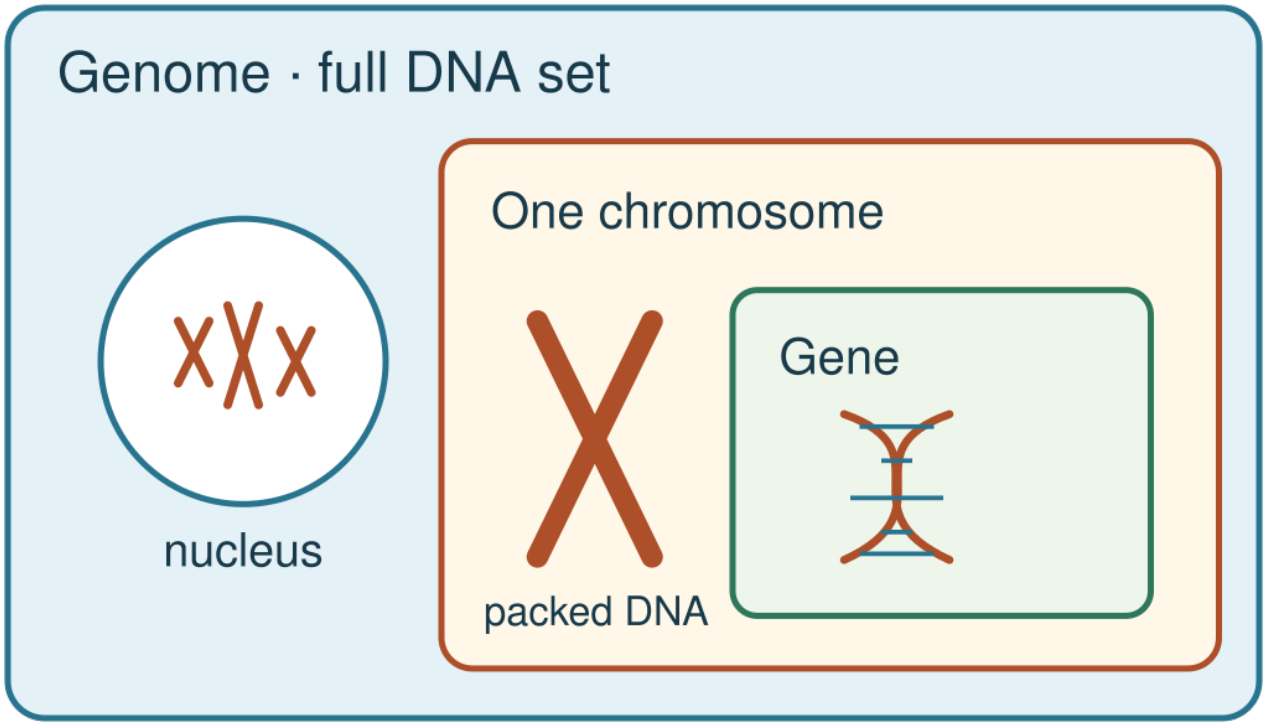
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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Arrival · Independent

Look at this week's string-and-clip DNA model. Point to one detail you notice.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Exit ticket · Independent

1. What does your model show, and why?

Show what you know with a model and explain one reason.

My learning: I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.

How I know: I can show a gene within DNA, DNA packaged in a chromosome, and the genome as all DNA.

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My feedback was … Name what you heard at Review.

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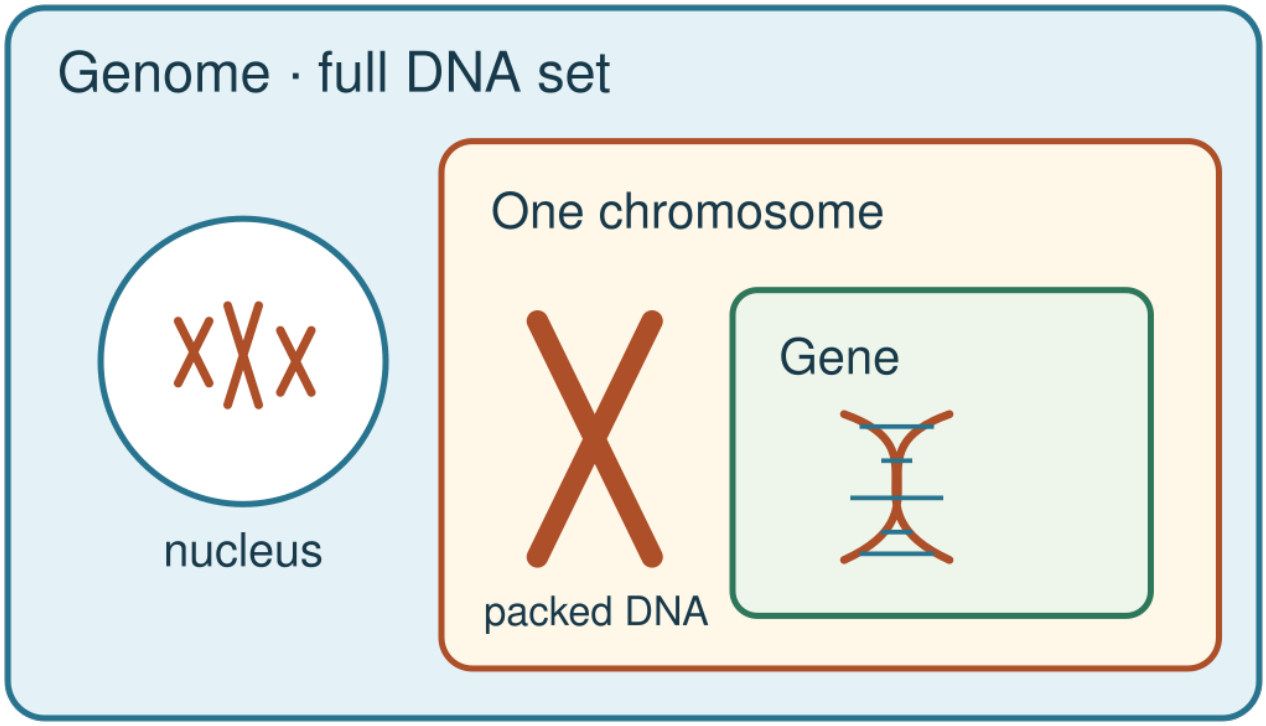
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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Arrival · Aspire

Look at this week's string-and-clip DNA model. Point to one detail you notice.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

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Exit ticket · Aspire

1. What does your model show, and why?

Show what you know with a model and explain one reason.

My learning: I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.

How I know: I can show a gene within DNA, DNA packaged in a chromosome, and the genome as all DNA.

2. What feedback did you hear, and what is your next step?

My feedback was … Name what you heard at Review.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Starter

Recall three ideas from earlier lessons. Show one answer at a time.

1. What is one reason a claim of a new stem-cell treatment needs evidence?
2. Which structure carries the matching chromosome sets after mitosis?
3. Where are chromosomes found in a eukaryotic animal cell?

Today I can:

I can label DNA, a gene and a chromosome in a simple model.

I can explain that a gene is a section of DNA with a protein instruction.

I can distinguish one chromosome from the genome.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

I do · words

Watch the full model. Point to the step that answers today's question.

Word	Meaning	Example
DNA	the molecule that stores genetic instructions	DNA contains the instructions for making proteins.
gene	a section of DNA that codes for a specific protein	The coloured section of the model stands for one gene.
chromosome	a long DNA molecule packaged with proteins in a cell	A chromosome contains many DNA sections, including genes.
genome	the entire DNA of an organism	The genome includes all of the organism's chromosomes and other DNA.

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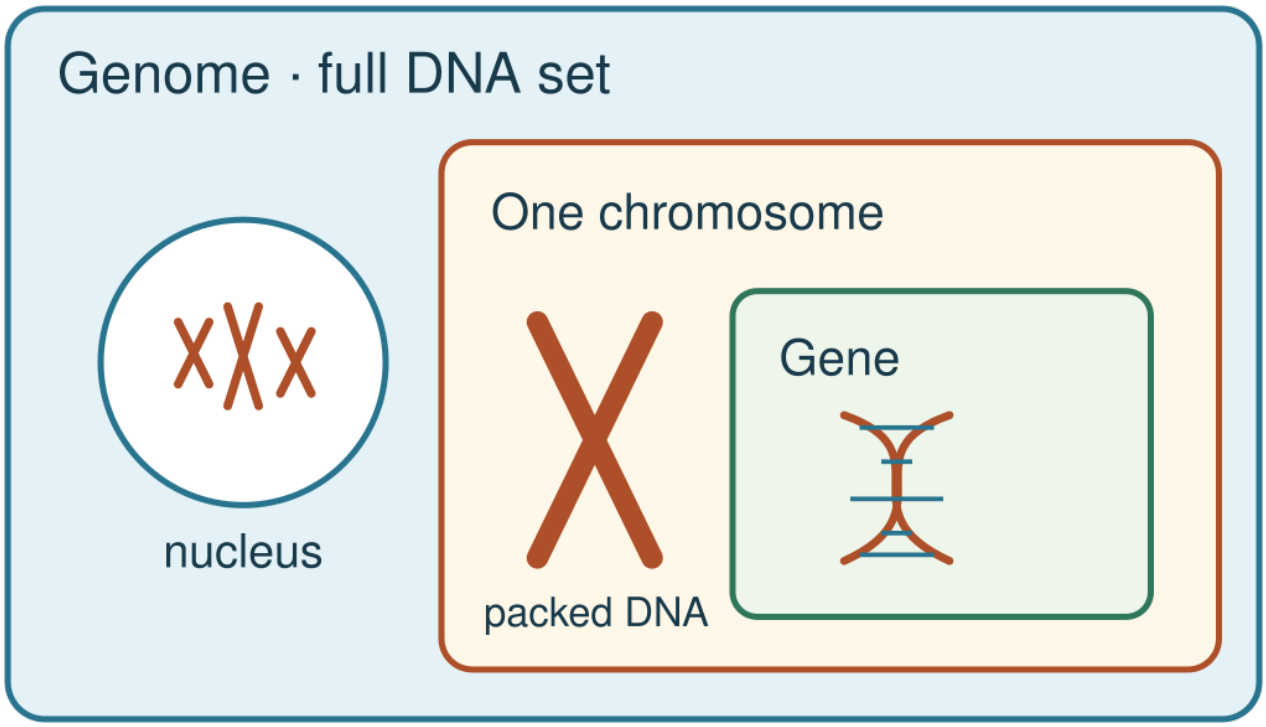
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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

I do · full model

Full model: Mark a short section of a DNA cord as a gene. Coil the cord as one chromosome; place it inside a border for the genome. Name the section, molecule, package and complete DNA set.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

Show me: is a gene a section of DNA, or is all DNA inside one gene?

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

We do · missing step

Complete the missing step in the part-done model. Speak, then match the cards.

One-minute reset: use the agreed move, then return to one step.

Part-done model: DNA cord → a short marked _____ codes for a protein → chromosome package → the genome contains all DNA.

Speak first: “A gene is part of …; a chromosome contains …”

Work as a pair. One explains a label. One asks which part it names. Swap roles. Each keeps their own response.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

We do · match the labels

Select where each label belongs in the string model and explain one mismatch.

Word bank, in source order: DNA / chromosome / gene / genome.

Clue	Your match
short coloured DNA section	Choose from the word bank.
whole genetic molecule	Choose from the word bank.
long DNA package	Choose from the word bank.
all of the organism’s DNA	Choose from the word bank.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

We do · choose a route

Supported: Use the full cord model. Place the gene label on the short marked section.

Independent: Explain how the short DNA section relates to one gene and the larger package.

Aspire: Explain why “one chromosome = one gene” misrepresents scale and role.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

We do 2 · hinge

Choose one hinge answer. Both learners show their answer at the same time.

A new diagram marks a short DNA region carrying instructions for one protein. Which label belongs to that region?

A. A genome containing the whole cell.

B. A gene within the DNA molecule.

C. A complete cell containing no DNA.

D. A chromosome containing the whole genome.

Show one answer to the adult. Point or use a card, too. Then explain to your partner. You may pass on speaking.

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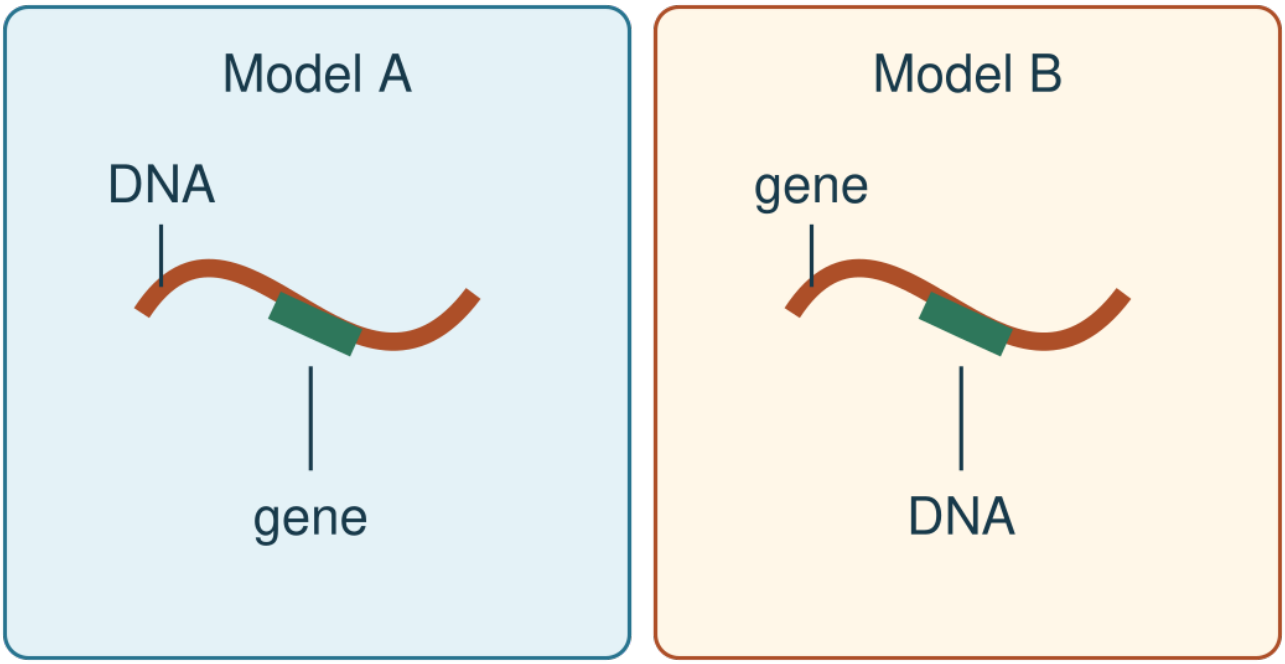
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Independent · compare the models

Check these two models. One has a false label. Choose the accurate model and explain one reason.



Both pictures are models to check, not real cells. Sizes are not to scale.

Talk before write: “Move this label because …”

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Independent · Supported

Two diagrams label a gene differently. Choose the accurate model and explain one reason.

One-minute reset: use the agreed move, then return to one step.

Talk before write: “Move this label because …”

You can plan in any language. Point, sign, speak, draw, type, write, use an exact scribe or a staff-captured photo.

Label a gene, DNA, chromosome and genome on the model.

This short marked section is a …; the long package is a …

Choose one route. Use the full 8 minutes for your own task.

I · S · C □

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Independent · Independent

Two diagrams label a gene differently. Choose the accurate model and explain one reason.

One-minute reset: use the agreed move, then return to one step.

Talk before write: “Move this label because …”

You can plan in any language. Point, sign, speak, draw, type, write, use an exact scribe or a staff-captured photo.

Explain why a gene is a DNA section, not another material.

A gene is part of … and codes for …

Choose one route. Use the full 8 minutes for your own task.

I · S · C □

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Independent · Aspire

Two diagrams label a gene differently. Choose the accurate model and explain one reason.

One-minute reset: use the agreed move, then return to one step.

Talk before write: “Move this label because …”

You can plan in any language. Point, sign, speak, draw, type, write, use an exact scribe or a staff-captured photo.

Correct a faulty gene label and explain the protein instruction.

The faulty label is …; it should be … because …

Choose one route. Use the full 8 minutes for your own task.

I · S · C □

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Independent · Go deeper

What distinctions disappear if the whole genome is called one gene?

Two-diagram audit: Repair two faulty diagrams: DNA inside a gene, and one chromosome as the entire genome. Justify both.

You can plan in any language. Point, sign, speak, draw, type, write, use an exact scribe or a staff-captured photo.

Choose one route. Use the full 8 minutes for your own task.

I · S · C □

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Independent · prepared model

This is model rehearsal, not a core practical.

Kit: Short DNA cord, large blunt clips, genome and chromosome frames, tray.

Mark one gene with a blunt clip and nest the cord model in its frames.

Risk: A clip could pinch a finger, and a loose cord could catch.

Safe: Use large blunt clips and short cord on a tray. Keep the cord away from necks.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Review and improve

Tell your partner what you changed after feedback. Mark a real edit // and read it back when asked ?.

One-minute reset: use the agreed move, then return to one step.

My feedback was … Name the actual point an adult or partner noticed. Make one true self-edit //. Ask “Read this back to me” ? if that helps.

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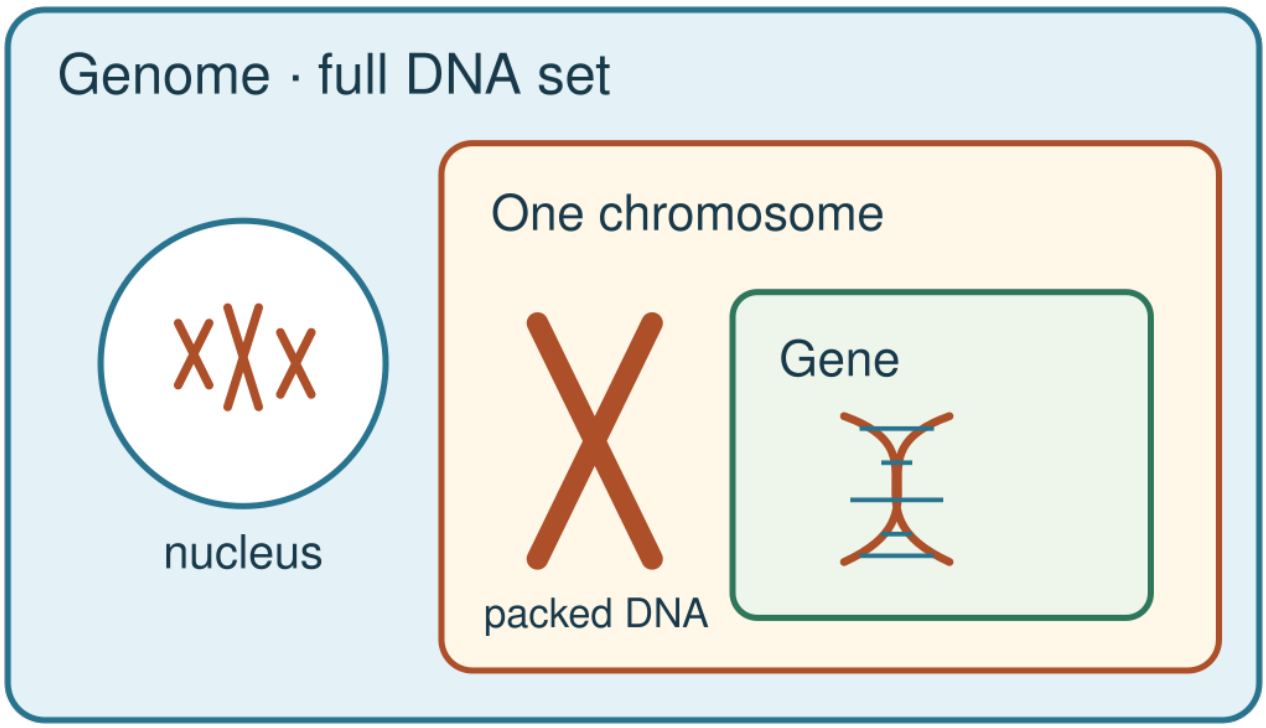
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Knowledge organiser · model

I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

Picture colours: blue genome frame, orange chromosome and DNA cord, green gene frame. Colours mark parts of the model; they are not real cell colours.

A gene is a DNA section with a protein instruction. One chromosome is not the whole genome.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Knowledge organiser · words

Word	Meaning
DNA	the molecule that stores genetic instructions
gene	a section of DNA that codes for a specific protein
chromosome	a long DNA molecule packaged with proteins in a cell
genome	the entire DNA of an organism

Mark a short DNA section as a gene. Pack the DNA cord as a chromosome. The genome is the whole DNA set, including other DNA.

No model shows the real sizes or all of a real cell. You may use words in any language to plan your explanation.

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Word page

Word	Meaning	Example
DNA	the molecule that stores genetic instructions	DNA contains the instructions for making proteins.
gene	a section of DNA that codes for a specific protein	The coloured section of the model stands for one gene.
chromosome	a long DNA molecule packaged with proteins in a cell	A chromosome contains many DNA sections, including genes.
genome	the entire DNA of an organism	The genome includes all of the organism’s chromosomes and other DNA.

My language: use words or notes that help you understand. Keep the science meaning.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Cards · match and explain

Word bank: DNA / chromosome / gene / genome.

Clue card
short coloured DNA section
whole genetic molecule
long DNA package
all of the organism’s DNA

Match each label and explain one mismatch.

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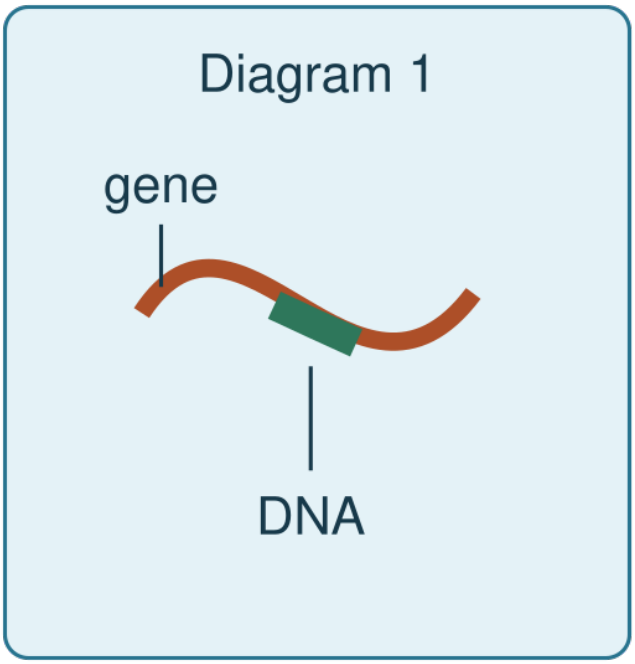
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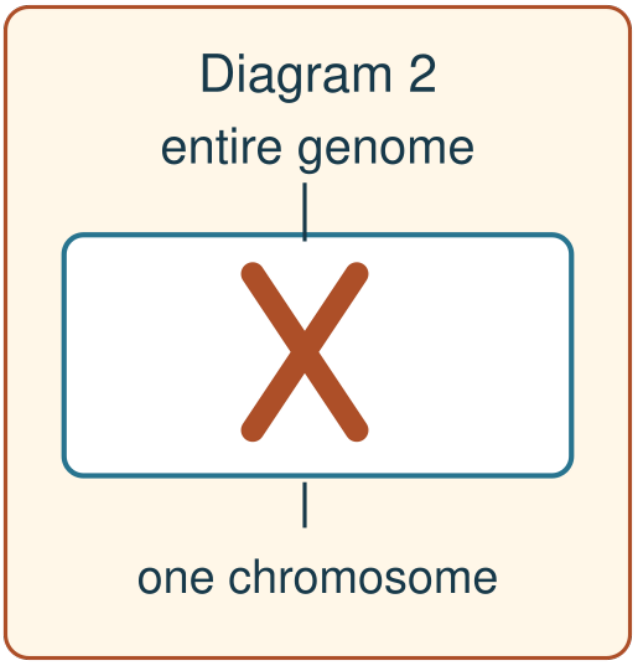
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Two-diagram audit · repair the faults

Both diagrams below contain a false label. Repair each one and justify both changes.





These are deliberately faulty challenge models. They are not teaching facts. Sizes are not to scale.

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